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UMBRELLA ASSEMBLY WITH COUNTERWEIGHT SYSTEM

Abstract

An umbrella is provided that includes a hollow pole, a shade assembly, a counterweight system, and a locking device. The shade assembly includes a guide rod coupled with an upper hub. The guide rod has a stop surface. The counterweight system has a ballast and a cord that has a first end coupled with the ballast and a second end coupled with a lower hub. The locking device has pushrod mechanism. When closing the umbrella, the pushrod mechanism moves upward disengaging a moveable locking surface from the guide rod to allow the lower hub to move downward and the ballast to move upward.

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Background/Summary

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS

[0001] Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 C.F.R. § 1.57.

BACKGROUND

Field of the Invention

[0002] This application is directed to umbrella assemblies having hub position locking systems and/or counterweight systems configured to assist the opening and/or closing of the umbrella and to ornamental designs of hubs and other components therefor.

Description of the Related Art

[0003] Many umbrellas are opened and closed by pushing a slider up and down a center pole of the umbrella. The length of the ribs of umbrellas usually depends on the size of the shade member. The larger the shade member, the longer the ribs are likely to be. Depending on the length of the ribs, the opening and closing of the umbrella can be interfered with by surrounding objects, such as tables and other furniture. This may impact the locations where the umbrellas can be used. Some umbrellas raise a top hub of the umbrella to try and avoid the surrounding objects. However, this may cause other issues such as instability due to wind.

SUMMARY

[0004] Instability due to wind points to a need for a locking device keeping a shade portion of the umbrella in an open or closed position. Such devices are important to provide a user with a stable umbrella that will not open or close independent of any action by the user. In view of the foregoing, improved umbrella assemblies incorporating counterweight systems are needed. Improved umbrella assemblies with locking devices for securing at least one of an open and a closed state are needed.

[0005] In various embodiments, an umbrella assembly is provided that includes a lower hub that can be moved up to open the umbrella and down to close the umbrella. The movement of the lower hub allows for movement of an upper hub that moves up when the umbrella is closed and down when the umbrella is opened. External factors, such as wind, can cause instability in the umbrella. Disclosed embodiments of this application encompass improved locking devices to better maintain the open position in a wide range of use conditions. Additionally, this invention increases the tolerance for the shade structure during production which is more conducive for mass production.

[0006] In one embodiment, an umbrella is provided that has a pole assembly, a shade assembly, a counterweight system, locking device, and a handle. The pole assembly has an upper portion and a lower portion. The pole assembly encloses an interior space disposed along a longitudinal axis thereof.

[0007] The shade assembly is disposed above the lower portion of the pole assembly and coupled to the upper portion of the pole assembly. The shade assembly has two or more ribs with a corresponding number of struts, a shade member, an upper hub, and a lower hub that is moveable along the pole assembly. The struts are coupled at one end to the ribs and are coupled at the other end to the lower hub. The upper hub includes a guide rod that is disposed within the pole assembly. The guide rod has one or more, e.g., at least two, recesses extending into the guide rod.

[0008] The counterweight system is disposed within the interior space of the pole assembly. The counterweight system has a ballast, a cord, and a pulley. The cord is coupled at a first end with the ballast and coupled at a second end with the lower hub. The cord is disposed over the pulley between the first and the second end of the cord.

[0009] The locking device has a hook assembly and a pushrod mechanism. The hook assembly can

be disposed on or mounted on the pole assembly. The hook assembly has one or more (e.g., two or at least two) prongs that are sized to fit within the one or more (e.g., two or at least two) recesses of the guide rod. The hook assembly can have a spring configured to urge the prongs into the one or more recesses to lock the umbrella in an open position. The pushrod mechanism has a fixture coupled with a lower end of the guide rod and a lower surface disposed opposite to the fixture. [0010] The handle extends downward from a lower portion of the lower hub. When opening the umbrella, the handle can be raised to raise the lower hub in the upward direction along the pole assembly as the ballast moves down within the interior space of the pole assembly. At the same time the upper hub is lowered until the one or more, e.g., at least two, recesses are aligned with the prong(s).

[0011] When closing the umbrella, the lower surface is actuated to move the fixture upward to disengage the prong(s) from the recess(es) of the guide rod. This allows the handle to move downward and the ballast to move upward.

[0012] In another embodiment, the fixture of the umbrella can further have an upper corner that is configured to disengage the prongs wherein the upper corner is angled.

[0013] In another embodiment, the fixture can have at least one width that corresponds to a width of an inner passage of the guide rod. The width of the fixture can be less than that of the fixture such that the fixture can slide within the inner passage. The width of the fixture can be sized relative to the passage to maintain the rotational position of the fixture within the passage.

[0014] In another embodiment, the pulley can be mounted to the upper portion of the pole.

[0015] In another embodiment, an umbrella is provided that has a pole assembly, a shade assembly, a counterweight system, and a locking device. The pole assembly encloses an interior space disposed along a longitudinal axis.

[0016] The shade assembly is coupled to an upper portion of the pole assembly. The shade assembly has a plurality of ribs, a plurality of struts, a shade member, an upper hub, and a lower hub. The struts are coupled at one end to a corresponding one of the ribs and at an opposite end to the lower hub. The upper hub has or is coupled with a guide rod disposed within the pole assembly. There is at least one recess extending into the guide rod.

[0017] The counterweight system is disposed within the interior space of the pole assembly. The counterweight system has a ballast and a cord. The cord has a first end coupled with the ballast and a second end coupled the lower hub. The locking device has a hook assembly and a pushrod mechanism. The hook assembly has at least one prong sized to fit within the at least one recess. A spring is optionally provided. Where provided, the spring can be configured to urge the at least one prong into engagement with the at least one recess to lock the umbrella in an open state. The pushrod mechanism has a pushrod and a fixture. The pushrod is coupled to the fixture. The fixture is coupled with a lower end of the guide rod.

[0018] When closing the umbrella, the pushrod is actuated to move the fixture upward to disengage the at least one prongs from the at least one recess of the guide rod. This allows the lower hub to be moved downward and the ballast to move upward.

[0019] In another embodiment, there may be a plurality of recesses extending into the guide rod. In another embodiment, the umbrella can further comprise a handle extending downward from a lower portion of the lower hub.

[0020] In another embodiment, the cord can be disposed over a pulley.

[0021] In another embodiment, when opening the umbrella, the lower hub can be raised in the upward direction along the pole assembly as the ballast moves down within the pole assembly, while at the same time the upper hub is lowered until the at least one recess is aligned with the at least one prong.

[0022] In another embodiment, the fixture can have an upper corner configured to disengage the at least one prong wherein the upper corner is angled.

[0023] In another embodiment, the fixture can have at least one width that corresponds to a width

of an inner passage of the guide rod. The width of the fixture can be less than that of the fixture such that the fixture can slide within the inner passage. The width of the fixture can be sized relative to the passage to maintain the rotational position of the fixture within the passage.

[0024] In another embodiment, an umbrella is provided that has a hollow pole, a shade assembly, a counterweight system, and a locking device. The shade assembly is coupled with an upper portion of the hollow pole. The shade assembly has an upper hub, a lower hub, a shade structure, a shade member, and a guide rod. The shade structure couples the upper hub to the lower hub. The shade structure supports the shade member. The lower hub is moveable along the hollow pole. The guide rod is coupled with the upper hub at a first end. A second end of the guide rod is disposed within the hollow pole. The guide rod has a stop surface.

[0025] The counterweight system is disposed within the hollow pole. The counterweight system has a ballast and a cord. The cord has a first end coupled with the ballast and a second end coupled with the lower hub.

[0026] At least a portion of the locking device is disposed on and/or in the hollow pole. The locking device has a moveable locking surface and a pushrod mechanism. The moveable locking surface can be coupled with the hollow pole. The moveable locking surface is sized to overlap with the stop surface to lock the umbrella in an open state. When closing the umbrella, the pushrod mechanism moves upward disengaging the moveable locking surface from the stop surface of the guide rod. This allows the lower hub to be moved downward and the ballast to move upward.

[0027] In another embodiment, the locking device further comprises a fixture coupled to the pushrod at an upper end thereof. The fixture is configured to urge the locking surface away from the stop surface.

[0028] In another embodiment, the moveable locking surface comprises a hook.

[0029] In another embodiment, the stop surface comprises a recess configured to engage the hook.

[0030] In another embodiment, the umbrella further comprises a fixture coupled to the pushrod mechanism wherein when closing the umbrella, the fixture disengages the moveable locking surface from the stop surface.

[0031] In another embodiment, the umbrella further comprises a hook pivotably mounted in the lower hub. The hook is configured to rest on a ledge of the pole assembly when the umbrella is open and to pivot away from the ledge to allow the umbrella to be closed. The pivoting of the hook is actuated by downward movement of a handle portion of the lower hub.

[0032] In another embodiment, the hook comprises a curved lower portion that is actuated by movement of a roller disposed in the handle.

[0033] In another embodiment, the cord is disposed over a pulley coupled with the pole assembly and is tension by downward movement of the lower hub.

[0034] In another embodiment, the umbrella further comprises a spring configured to engage the moveable locking surface with the stop surface.

[0035] In another embodiment, the fixture has an upper corner configured to disengage the at least two prongs wherein the upper corner is angled.

[0036] In another embodiment, the upper corner comprises a wedge surface configured to act on a wedge surface of the hook.

[0037] In another embodiment, downward movement of the lower hub raises the ballast into engagement with an extender coupled with the guide rod and further downward movement of the lower hub causes the guide rod to apply an upward force to the extender and thereby to the guide rod to raise the upper hub.

[0038] In another embodiment, the umbrella further comprises an upper spacer coupled with a lower portion of the guide rod and a lower spacer disposed below the upper spacer. The upper and lower spacers are disposed in the pole assembly. The pushrod mechanism is moveable through the spacers between a position below the locking surface to a position at or above the locking surface.

[0039] In another embodiment, the pushrod mechanism includes a fixture coupled with an upper

portion of a pushrod. The pushrod has a slot disposed therethrough. The slot can at least partially limit the range of motion of the fixture.

[0040] In another embodiment, the umbrella further comprises a guide rod lifting mechanism comprising a second cord coupled at a first end and at a second end opposite the first end with an upper portion of the hollow pole. The second cord having a span extending from the first end to a second end. The span of the second cord is disposed over a pulley. The span of the second cord being coupled with an upper end of an extender within the guide rod, wherein downward movement of the lower hub causes upward movement of the extender, whereby the length of a portion of the span of the second cord between the pulley and the first end of the cord is shortened and the guide rod and the upper hub are raised.

[0041] In another embodiment, the guide rod lifting mechanism comprises a second pulley. The span of the second cord extends between the first end and the first pulley, over the upper end of the extender, and from the second pulley to the second end, wherein the length of a portion of the span of the second cord between the second pulley and the second end of the cord is shortened when the lower hub is moved downward.

[0042] In some embodiments, an umbrella hub as shown and described is provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0043] Features of the invention can be better understood from the following detailed description when read in conjunction with the accompanying schematic drawings, which are for illustrative purposes only. The drawings include the following figures:

[0044] FIG. 1 shows an umbrella including a pole assembly, a shade assembly, an upper hub, a lower hub and a counterweight assembly;

[0045] FIG. 2 shows the umbrella of FIG. 1 in a closed state where the upper hub has moved upwards compared to the position shown in FIG. 1;

[0046] FIG. 3 shows one embodiment of an improvement of an umbrella similar to that of FIG. 1;

[0047] FIG. 4 shows a lower hub, a handle, and a covered locking device of the umbrella of FIG. 3;

[0048] FIG. 4A shows a cross section of the umbrella FIG. 4 showing features of a counterweight assembly and a locking device including a hook, a pushrod mechanism, a pulley, and a cord;

[0049] FIG. 5 shows a cross section of the umbrella FIG. 4 illustrating a central portion of a pole assembly including a pushrod;

[0050] FIG. 6 shows a cross section of the umbrella FIG. 4 illustrating a pole assembly including a ballast connected to a cord and a bottom portion of a push rod;

[0051] FIG. 6A shows a top perspective view of portion of a push rod and a ballast disposed within a lower portion of a pole assembly, an outer pole portion shown in phantom;

[0052] FIG. 6B shows a cross section of the umbrella of FIG. 4 illustrating a guide rod lifting mechanism including a pulley, a cord, and an extender;

[0053] FIG. 6C shows a cross section of the umbrella of FIG. 4 illustrating an upper hub and the cord illustrated in FIG. 6B;

[0054] FIG. 7 shows a cross section of the umbrella of FIG. 4 illustrating a lower hub and a handle when an umbrella is in an open state;

[0055] FIG. 8 shows a cross section of the umbrella of FIG. 4 similar to that of

[0056] FIG. 7 illustrating a lower hub and a handle when a handle has been pulled down to disengage the lower hub from a ledge securing the lower hub to the pole assembly;

[0057] FIG. 9 shows a top perspective view of a portion of a lower hub assembly, an upper portion thereof having been removed to show a spacer;

[0058] FIG. 10 shows a perspective view of a design of an umbrella hub with a handle;

[0059] FIG. **11** shows a top view of the umbrella hub design of FIG. **10**;

[0060] FIG. **12** shows a bottom view of the umbrella hub design of FIG. **10**;

[0061] FIG. **13** shows a side view of the umbrella hub design of FIG. **10** with the handle in a first position; and

[0062] FIG. **14** shows a side view of the umbrella hub design of FIG. **10** with the handle in a second position.

DETAILED DESCRIPTION

[0063] While the present description sets forth specific details of various embodiments, it will be appreciated that the description is illustrative only and should not be construed in any way as limiting. Furthermore, various applications of such embodiments and modifications thereto, which may occur to those who are skilled in the art, are also encompassed by the general concepts described herein. Each and every feature described herein, and each and every combination of two or more of such features, is included within the scope of the present invention provided that the features included in such a combination are not mutually inconsistent.

[0064] This application is directed to new umbrella assemblies that facilitate one or both of counterweight assist in the opening and/or closing of large umbrellas and position locking of one or more hubs of an umbrella to enhance stability of the umbrella in an open or closed state. These assemblies can be embodied in or operate with novel hub configurations and designs. A combination of one or more, e.g., one or more pairs, e.g., two pairs of pulleys on or within a pole assembly facilitate movement of the upper and lower hubs as well as a ballast of a counterweight system.

[0065] FIGS. **1** and **2** show an umbrella assembly **50** including a pole assembly **54**, a guide tube **56**, a shade assembly **58**, an upper hub **62**, a lower hub **64**, a ballast **78**, and a pulley **82**. The shade assembly **58** includes ribs **66**, struts **70**, and a shade member (not shown). The ribs **70** are coupled to the upper hub **62** along their length to one end of corresponding struts **66** at the other end. The end of the struts **66** that are not coupled to the ribs **70** are coupled to the lower hub **64**. When the umbrella assembly **50** closes the lower hub **64** is lowered and the upper hub **62** is raised as shown in FIG. **2**. The raising of the upper hub **62** also elevates the ribs **70**. If the upper hub **62** were not enabled to be raised during closing of the umbrella the lower end of the ribs **70** would swing down much lower than is seen in FIG. **2**. Thus, the elevation of the upper hub **62** while closing prevents the long ribs **70** from coming in contact with other objects, such as tables or chairs, when closing.

[0066] FIG. **3** shows an example improved umbrella **100** including a pole assembly **104** and a shade assembly **108**, the shade assembly **108** including an upper hub **120** and a lower hub **124**. The umbrella **100** also includes a counterweight assembly **112** in some embodiments, for example as shown in FIG. **6A**. The umbrella **100** also includes a locking device **116** in some embodiments, for example, as shown in FIG. **4A**. The pole assembly **104** can include an upper portion **126** and a lower portion **130**. The pole assembly **104** can enclose an interior space along a central longitudinal axis. The shade assembly **108** can be disposed above the lower portion **130** of the pole assembly **104**. The shade assembly **108** includes struts **134** and ribs **138**. There can be any number of struts **134** and ribs **138**, for example three, four, five, six, or more ribs. The struts **134** can be coupled to ribs **138** at or near a first end and coupled to the lower hub **124** at a second end. The ribs **138** can be coupled to the upper hub **120** at or near a first end and can have a second end opposite the first end connected to a shade member **142** (shown in dashed lines) of the shade assembly **108**. The upper hub **120** can move upwards when the umbrella **100** is closed (e.g., as shown in FIG. **2**).

[0067] The lower hub **124** can be disposed on the pole assembly **104**. The lower hub **124** can be moveable along the pole assembly **104**. For example, the lower hub **124** can be moved in the downward direction to close the umbrella **100** and moved in the upward direction to open the umbrella **100**. The lower hub **124** can include a handle **125** disposed on the pole assembly **104**. In some embodiments, the handle **125** can be pulled down and pushed up to move the lower hub **124** along the pole assembly **104**. The handle **125** can also function to disengage a locking mechanism

of the lower hub **124** as discussed further below. In some embodiments a separate handle **125** may not be needed and a lower hub **124** can function as both a handle and lower hub. The upper hub **120** can be disposed at least partially at or near the upper portion **126** of the pole assembly **104**. The upper hub **120** can include a guide rod **121** that is moveable within the pole assembly **104**. The guide rod **121** can include an inside passage **122** defined within a hollow space of the guide rod. [0068] FIGS. **4-6A** show further details of the counterweight assembly **112** and the locking device **116**. The locking device **116** works with the counterweight assembly **112** in some embodiments to enable the user to manipulate the umbrella **100** between open and closed states and also to provide for securing the umbrella **100** in one of these states, e.g., in the open state. In some embodiments, the locking device **116** can be used without a counterweight assembly **112**.

[0069] The counterweight assembly **112** can include a ballast **170** (e.g., as shown in FIG. **6**), a pulley **178** (e.g., as shown in FIG. **4A**), and a cord **180** (e.g., as shown in FIG. **4A**, **5** and **6**). The counterweight assembly **112** can be disposed within the interior of the pole assembly **104**. As shown in FIG. **4A**, two pulleys **178** can be located along the pole assembly **104** above the lower hub **124**. The cord **180** can be attached to the lower hub **124** by any suitable means. For example, one or more ends of the cord **180** can be anchored in a passage of an upper portion of the lower hub **124**. The cord **180** can be positioned over a first pulley **178** and can extend down the interior of the pole assembly **104**. In one embodiment, a space **181** is defined within the interior of the pole assembly **104** between, at, or near the lower hub **124** and the lower end **143** of the pole assembly **104**. The space **181** can be hollow and the cord **180** can be positioned within the space **181**. As shown in FIG. **5**, the cord **180** extends between an outer surface of the guide rod **121** and the inner wall **105** of the pole assembly **104**. In some embodiments, the guide rod **121** does not extend to the lower end **143** of the pole assembly **104**. As shown in FIG. **5**, the guide rod **121** can end or have a lowermost portion at an intermediate portion **226** of the pole assembly **104**. The cord **180** can pass through two openings **220** when extending through space **181**. The openings **220** can be disposed through an upper spacer **183** coupled with a lower end of the guide rod **121**. The upper spacer **183** can help to maintain the guide rod **121** in alignment within the pole assembly **104**.

[0070] As shown in FIGS. **6** and **6A**, the cord **180** can continue to extend through space **181**. The cord **180** can pass through two spacer openings **224** of a lower spacer **216**. The cord **180** can be attached to the ballast **170** and the remaining portion of the cord **180** can extend up the interior of the pole assembly **104**, as previously described, and can be positioned over a second pulley **178** and attached to the lower hub **124** by any suitable means. In one embodiment, the ballast **170** includes a channel **171** and the cord **180** is passed or threaded through the channel **171**, as shown in FIG. **6**. In this approach a single cord can extend from the first pulley **178** to the second pulley **178**.

[0071] When the umbrella is opened, the handle **125** can be pushed upwards moving the lower hub **124** in the upward direction as the length of the cord **180** disposed between the pulley **178** and the lower hub **124** becomes less as more of the cord **180** is disposed between the pulley **178** and the ballast **170**. The movement of the lower hub **124** upward will move the ballast **170** downward within the interior of the pole assembly **104**. When the umbrella is closed, the handle **125** can be pulled downward which will move the lower hub **124** downward and the ballast **170** will move upward. This movement will cause more length of the cord **180** to be disposed between the pulley **178** and the lower hub **124**. Stated another way, the ballast **170** can be configured as a weight that unless restrained will be guided by gravity downward within the space **181** of the pole assembly **104**. The downward movement of the ballast **170** will raise or will help to raise the lower hub **124** which will cause the struts **134** and the ribs **138** (together a frame) of the shade assembly **108** to open.

[0072] Although the ballast **170** tends to hold the umbrella **100** in an open configuration absent a user collapsing it, the open position can be secured by one or more locking devices. As discussed above, the locking device **116** secures the elevation of the guide rod **121** relative to the outer pole of pole assembly **104**. Also, as discussed further below, one or more hooks **234** disposed within the

lower hub **124** can secure the elevation of the lower hub **124** relative to the pole assembly **104**. [0073] A pushrod **190** and a hook assembly **154** (discussed below) can form parts of the locking device **116**. The hook assembly **154** can lock the position of the upper hub **120** relative to the pole assembly **104** in an open state. The pushrod **190** can be used to disengage the hook assembly **154** as discussed below. In one convenient configuration the pushrod **190** is operated by movement of the ballast **170** within the space **181** disposed within the pole assembly **104**. The distance the pushrod **190** can be moved by the ballast **170** in order to push the pushrod **190** up to the hook assembly **154** can be predetermined. As discussed further below, a wedge element can be used to urge prongs **158** of the hook assembly **154** out of recesses **162** formed in or through the guide rod **121**. For example, in some embodiments the ballast **170** can move the pushrod **190** upward a distance **L** (see FIG. **6**). The distance **L** can be sufficient for the pushrod **190** to disengage the locking device **116**, as described herein. As shown in FIG. **6A**, a slot **212** positioned in the pushrod **190** can be configured to limit the distance **L** that the pushrod **190** can move relative to a lower spacer **216**. As described above, the ballast **170** will move the pushrod **190** in the upward direction. When the pushrod **190** has moved a distance **L**, the slot **212** and a rod or pin **230** extending from the lower spacer **216** through the lower spacer **216** will prevent further movement of the pushrod **190**. The pin **230** can extend through the slot **212** and into the lower spacer **216** to limit the movement of the pushrod **190**. The bottom **194** of the pushrod **190** will now be at or near the bottom of the lower spacer **216**.

[0074] Limiting movement can be beneficial because it limits the amount of movement necessary to disengage the locking device **116** to allow the umbrella **100** to close when in an open configuration, therefore reducing potential problems due to internal movement of parts. In some embodiments, the ballast **170** continues to move in the upward direction once the pushrod **190** has moved a distance **L**. The ballast **170** will continue to move in the upward direction which will move the guide rod **121** in the upward direction. In some embodiments, the upward movement of the guide rod **121** can cause the upper hub **120** to move upwards when the umbrella is closing as described herein. In one embodiment, the guide rod **121** is above the ballast and an extender **121A** may be provided between a lower end of the guide rod **121** and the ballast **170**. The extender **121A** can be a member configured to support a compressive load such that as an upward force is applied by the ballast **170** to the lower spacer **216**, that load is transferred to the extender **121A** and is transferred by the extender **121A** to the guide rod **121**.

[0075] The hook assembly **154** can be coupled with the intermediate portion **167** of the pole assembly **154**. The hook assembly **154** can be covered by a cover **204** to protect the hook assembly **154** from external elements. The hook assembly **154** can include prongs **158**. Any number of prongs **158** can be used. For example, one, two, three, or more, prongs **158** can be used. The lowest of the prongs **158** can have an angled lower portion **159** that will engage with the fixture **186** described herein. The prongs **158** are configured to fit into corresponding recesses **162** in or through the guide rod **121**. The corresponding recesses **162** can have sufficient space to allow the prongs **158** to engage and disengage from the recesses **162**. In the illustrated embodiment, the recesses **162** extend through guide rod **121** but the recesses do not have to extend all the way through the guide rod **121**. The guide rod **121** can be positioned in and moveable within the pole assembly **104**. The prongs **158** can be urged into the corresponding recesses **162** by a spring **166** to lock the umbrella in an open state. Any suitable spring **166**, e.g., a coil spring, a torsion spring, a leaf spring, can be used.

[0076] The pushrod **190** can be part of a pushrod mechanism **174** (e.g., as shown in FIG. **4A**). The pushrod mechanism **174** can include a fixture **186** that can be coupled to the guide rod **121** and coupled to side surface **191** of the pushrod **190**. The fixture **186** is configured to disengage the prongs **158** from the corresponding recesses **162** when the umbrella is closed. The fixture **186** can be any suitable shape. In one embodiment, the fixture can be at least partially wedge-shaped. An angled upper corner **208** of the fixture **186** can form a portion of a wedge that is configured to have

disengage the prongs **158** from the recesses **162**. The angled upper corner **208** can be positioned over the pushrod **190** as shown in FIG. **4A**. The fixture **186** has an outer periphery **187** that has a square or rectangular shape. The shape of the outer periphery **187** can be the same as the inner walls of the guide rod **121**. The width of the fixture **186** corresponds to the width of the inside passage **122** of the guide rod **121**. The corresponding widths and the outer periphery **187** matching up with the inner walls of the guide rod **121** allows the fixture **186** to engage with the prongs **158** of the hook assembly **154**. The corresponding widths will prevent or limit the fixture **186** from deflecting away from the prongs **158** when the fixture **186** pushes against the angled lower portion **159** of the prongs **158**.

[0077] When the umbrella is opened the ballast **170** moves downward. The ballast **170** moving downward allows for the pushrod **190** to move downward and the fixture **186** to move downward. The fixture **186** moving downward can allow the spring **166** of the hook assembly **154** to urge the prongs **158** into the corresponding recesses **162** to lock the umbrella in an open state. The use of two prongs **158** can be beneficial to increase the stability of the umbrella **100** in a locked or open state.

[0078] As described herein, when the umbrella **100** is to be closed the ballast **170** is moved upward. The ballast **170** moving upward pushes the pushrod **190** upward which advances the fixture **186** upward toward the prongs **158**. When the angled upper corner **208** of the fixture **186** contacts the angled lower portion **159** of the prongs **158**, the fixture **186** disengages the prongs **158** from the corresponding recesses **162**. Where the fixture **186** and the lower prong **158** have an angled shape, the prongs **158** more easily disengaging from the corresponding recesses **162**. The angled upper corner **208** of the fixture **186** removes the risk of the fixture **186** not disengaging the prongs **158**. For example, if the engagement portion of the fixture **186** was square or rectangular shape the disengagement may not be as smooth if the fixture were to get stuck when moving up to disengage the prongs **158**. After disengaging the prongs **158** from the corresponding recesses **162**, the guide rod **121** can move up which can move the upper hub **120** upwards. As described herein, the upper hub **120** moving upwards when the umbrella **100** closes can prevent or limit the ribs **138** from contacting other objects external to the umbrella **100**, such as tables or chairs.

[0079] FIGS. **6B** and **6C** illustrate section views of the umbrella **100** having a guide rod lifting assembly **308**. The guide rod lifting assembly **308** can move the upper hub **120** upward when the umbrella **100** is closed. The guide rod lifting assembly **308** can include a cord **310**, at least one pulley **314**, and an extender **121A**. As described above, the extender **121A** can be disposed within the pole assembly **104** and can be positioned next to the pushrod **190**. At least one pulley **314** is coupled to the upper spacer **183** cap disposed at a bottom portion of the guide rod **121**. The extender **121A** can be slideably disposed through an aperture in the upper spacer **183** such that a top portion thereof can be extended past the elevation of the pulley(s) **314** as discussed below. The extender **121A** can have a fixture **334** coupled to the upper portion of the extender **121A**, as shown in FIGS. **5** and **6B**. The fixture **334** can have an opening **338** that the cord **310** can extend through. The lower portion **342** of the opening **338** may have a rounded edge, as shown in FIG. **6B**.

[0080] The pole assembly **104** can have ring **326** coupled to the upper portion of the pole assembly, as shown in FIG. **6C**. The ring **326** can have points **330a**, **330b** where the cord **310** can be coupled at each end to the pole assembly **104** by any suitable means. A first end of the cord **310** can be coupled or anchored to point **330a**. The cord can then extend down the pole assembly **104** at least partially between an inner surface of the pole assembly **104** and an outer surface of the guide rod **121**. The cord **310** can then extend under a pulley **314** upward and through the opening **338** of the fixture **334**. The lower portion **342** of the opening **338** can have a rounded edge or profile that can allow the cord **310** to extend through the opening **338** without any sharp turns or sharp edges. The cord **310** can then extend downward from the fixture **334** and under a second pulley **314**. The cord **310** can then extend upward between an inner surface of the pole assembly **104** and an outer surface of the guide rod **121**. The second end of the cord **310** can be coupled or anchored to point

330b. In one configuration, the points **330a**, **330b** are openings in an upper surface of the ring **326**. The cord **310** can be knotted at an end portion above the openings in the ring **326** such that the cord **310** will not pull through the openings.

[0081] As described throughout, the upper hub **120** of the umbrella **100** can move in the upward direction when the umbrella **100** is closed. The lower hub **124** can be moved downward to close the umbrella **100**. As described in more detail above, when the lower hub **124** is moved downward, the ballast **170** moves upward pushing the pushrod **190** upward which advances the fixture **186** upward toward the prongs **158** to unlock the umbrella **100**. When the pushrod **190** has moved a distance **L**, as described above, the ballast **170** moving upward can continue to move the extender **121A** in the upward direction. The extender **121A** can comprise a second pushrod disposed alongside the pushrod **190**. The extender **121A** moving upward can apply a force to the guide rod **121** and assist the guide rod **121** to move upward, which can assist the upper hub **120** in moving upward. In particular, the upward movement of the extender **121A** raises the span of the cord **310** over the lower portion **342** of the opening **338**. The ends of the cord **310** at points **330a**, **330b** are fixed. Raising the span of the cord **310** over the lower portion **342** raises the pulleys **314** relative to the points **330a**, **330b**, which raises the guide rod **121**. While the extender **121A** is moving upward (e.g., when closing the umbrella), the length of the cord **310** between points **330a**, **330b** and the pulleys **314** will decrease, while the length of the cord **310** between the opening **338** and the pulleys **314** will increase. When the extender **121A** moves downward (e.g., when opening the umbrella), the length of the cord **310** between points **330a**, **330b** will increase, while the length of the cord **310** between the opening **338** and the pulleys **314** will decrease. Also, while the extender **121A** is moving upward, the length of the extender **121A** within the guide rod **121** will increase. When the extender **121A** moves downward (e.g., when opening the umbrella), the length of the extender **121A** within the guide rod **121** will decrease.

[0082] FIG. 7 illustrates a section view of the lower hub **124** and the handle **125** of an umbrella **100** when the umbrella **100** is in an open state. The handle **125** is positioned in a recess **233** of the lower hub **124**. When the handle **125** is pulled down, the handle **125** can create a separation between the upper portion **250** of the handle **125** and an internal aspect of the recess **233** of the lower hub **124**. The lower hub **124** is secured to the pole assembly **104** via hooks **234**. The illustrated embodiment shows two hooks **234**, but any number of hooks **234** can be used. For example, 1, 2, 3, 4, or more hooks **234** can be used. The hooks **234** can include a lower rounded portion **258** and a prong **262** configured to engage a ring **242**. The ring **242** can be coupled to the exterior of the pole assembly **104**. The ring **242** can include a ledge **246**.

[0083] The hooks **234** engage with ring **242** to secure the lower hub **124** in a locked state, securing the umbrella **100** in an open state. In one embodiment, the hooks **234** can be used with the locking device **116**, described above, to better secure the umbrella **100** in an open state. The ring **242** partially or completely encircles the pole assembly **104**. The ledge **246** can engage the hooks **234**. The ledge **246** can extend around the entire ring **242** or one or more ledges **246** can be provided at the location of the hooks **234**. The lower rounded portion **258** of the hooks **234** are configured to interact with pins **254** to secure and unsecure the lower hub **124** to the pole assembly **104**.

[0084] FIG. 8 illustrates a section view of a lower hub **124** and a handle **125** of an umbrella **100** when the umbrella **100** is being moved to a closed state. To close the umbrella **100**, the handle **125** can be pulled down relative to the portion of the lower hub **124** coupled with the struts **134**. The upper portion **250** of the handle **125** moves downward from the recess **233** of the lower hub **124** as shown in FIGS. 7 and 8. The downward movement of the upper portion **250** of the handle **125** can allow pins **254** to move down and apply a force to the lower rounded portion **258** of the hooks **234**. The application of this force can cause the hooks **234** to disengage from the ledge **246** of the ring **242**. FIG. 8 illustrates the hooks **234** moving away from the ledge **246** of the ring **242**. The distance the hooks **234** move from the ledge **246** of the ring **242** can be limited by an interior wall **266** of the lower hub **124**. The interior wall **266** encircles the hooks **234** around the pole assembly **104**. The

interior wall **266** limiting the distance the hooks **234** can allow for a more controlled locking and unlocking process. The disengagement of the hooks **234** can allow the lower hub **124** to move down the pole assembly **104** allowing the umbrella **100** to close.

[0085] FIG. **9** shows a top perspective view of a bottom portion **270** of the lower hub **124**. An upper portion (not shown) has been removed to illustrate internal components. A spacer **274** is positioned in the center of the bottom portion **270** between the pole assembly **104** and the lower hub **124**. The spacer **274** can be made of plastic. In the illustrated embodiment, the spacer **274** includes aligners **278**. There can be one, two, three, four or more aligners **278**. The aligners **278** can be used to secure the spacer **274** in place or in a predetermined location. In the illustrated embodiment, the aligners **278** are ring-like tabs securing the spacer **274** to the lower hub. The ring-like tabs of the aligners **278** can be advanced over pins molded, formed or mounted in the bottom portion **270** of the lower hub **124**. In other embodiments, the aligners **278** or other suitable means can be provided to couple or attach the spacer **274** to the lower hub **124**.

[0086] In the illustrated embodiment, two spacers **274** can be provided. In other embodiments, the spacer **274** can be one piece the encircles the entire diameter of the pole assembly **104**. In some embodiments, the pole assembly **104** and the lower hub **124** can be metallic. The lower hub **124** moves along the pole assembly **104** as previously described and the spacer **274** can prevent or limit the wearing down of the pole assembly **104** and the lower hub **124**.

[0087] FIGS. **10-14** show a design of an umbrella hub **290**. FIG. **10** shows a perspective view of a design of an umbrella hub **290** with a handle **294** in a first position. An upper portion of the umbrella hub **290** can be positioned above the handle **294**. An upper portion of the handle **294** is positioned in a recess **298** of the umbrella hub **290**. The upper portion of the umbrella hub **290** can include two portions, a top portion **302** and a bottom portion **306**. FIG. **11** shows a top view of the top portion **302** of the umbrella hub **290** design of FIG. **10**. FIG. **12** shows a bottom view of the bottom portion **306** and of the handle **294** of the umbrella hub **290** design of FIG. **10**.

[0088] FIG. **13** shows a side view of the umbrella hub **290** design of FIG. **10** with a handle **294** in a first position. The handle **294** is positioned in the recess **298** of the umbrella hub **290**. FIG. **14** shows a side view of the umbrella hub **290** design of FIG. **10** with a handle **294** in a second position. In FIG. **14**, an upper portion of the handle **294** has been moved downward and partially out of the recess **298**.

[0089] While certain embodiments of the inventions have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms. Furthermore, various omissions, substitutions and changes in the systems and methods described herein may be made without departing from the spirit of the disclosure. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure. Accordingly, the scope of the present inventions is defined only by reference to the appended claims.

[0090] Features, materials, characteristics, or groups described in conjunction with a particular aspect, embodiment, or example are to be understood to be applicable to any other aspect, embodiment or example described in this section or elsewhere in this specification unless incompatible therewith. All of the features disclosed in this specification (including any accompanying claims, abstract and drawings), and/or all of the steps of any method or process so disclosed, may be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive. The protection is not restricted to the details of any foregoing embodiments. The protection extends to any novel one, or any novel combination, of the features disclosed in this specification (including any accompanying claims, abstract and drawings), or to any novel one, or any novel combination, of the steps of any method or process so disclosed.

[0091] Furthermore, certain features that are described in this disclosure in the context of separate

implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations, one or more features from a claimed combination can, in some cases, be excised from the combination, and the combination may be claimed as a subcombination or variation of a subcombination.

[0092] Moreover, while operations may be depicted in the drawings or described in the specification in a particular order, such operations need not be performed in the particular order shown or in sequential order, or that all operations be performed, to achieve desirable results. Other operations that are not depicted or described can be incorporated in the example methods and processes. For example, one or more additional operations can be performed before, after, simultaneously, or between any of the described operations. Further, the operations may be rearranged or reordered in other implementations. Those skilled in the art will appreciate that in some embodiments, the actual steps taken in the processes illustrated and/or disclosed may differ from those shown in the figures. Depending on the embodiment, certain of the steps described above may be removed, others may be added. Furthermore, the features and attributes of the specific embodiments disclosed above may be combined in different ways to form additional embodiments, all of which fall within the scope of the present disclosure. Also, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described components and systems can generally be integrated together in a single product or packaged into multiple products.

[0093] For purposes of this disclosure, certain aspects, advantages, and novel features are described herein. Not necessarily all such advantages may be achieved in accordance with any particular embodiment. Thus, for example, those skilled in the art will recognize that the disclosure may be embodied or carried out in a manner that achieves one advantage or a group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

[0094] Conditional language, such as “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements, and/or steps are included or are to be performed in any particular embodiment.

[0095] Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require the presence of at least one of X, at least one of Y, and at least one of Z.

[0096] Language of degree used herein, such as the terms “approximately,” “about,” “generally,” and “substantially” as used herein represent a value, amount, or characteristic close to the stated value, amount, or characteristic that still performs a desired function or achieves a desired result. For example, the terms “approximately,” “about,” “generally,” and “substantially” may refer to an amount that is within less than 10% of, within less than 5% of, within less than 1% of, within less than 0.1% of, and within less than 0.01% of the stated amount. As another example, in certain embodiments, the terms “generally parallel” and “substantially parallel” refer to a value, amount, or characteristic that departs from exactly parallel by less than or equal to 15 degrees, 10 degrees, 5 degrees, 3 degrees, 1 degree, or 0.1 degree.

[0097] The scope of the present disclosure is not intended to be limited by the specific disclosures

of preferred embodiments in this section or elsewhere in this specification, and may be defined by claims as presented in this section or elsewhere in this specification or as presented in the future. The language of the claims is to be interpreted broadly based on the language employed in the claims and not limited to the examples described in the present specification or during the prosecution of the application, which examples are to be construed as non-exclusive.

Claims

1.-28. (canceled)

29. An umbrella comprising: a pole; a shade assembly coupled with an upper portion of the pole, the shade assembly comprising an upper hub; a guide rod movably disposed within the pole and coupled with the upper hub; and a locking device, the locking device comprising: a hook assembly configured to engage the guide rod to lock the umbrella in an open state, and a pushrod at least partially disposed below the hook assembly, wherein, when closing the umbrella, the pushrod is actuated to move upward to disengage the hook assembly.

29. The umbrella of claim **29**, wherein the hook assembly comprises a prong configured to fit within a corresponding recess of the guide rod.

31. The umbrella of claim **30**, wherein the hook assembly comprises a plurality of prongs, each prong configured to fit within a corresponding recess of the guide rod.

32. The umbrella of claim **29**, wherein the pushrod comprises a fixture configured to disengage the hook assembly when the pushrod moves upward.

33. The umbrella of claim **32**, wherein the fixture comprises an angled surface, the angled surface configured to disengage the hook assembly when the pushrod moves upward.

34. The umbrella of claim **29**, wherein the pushrod comprises at least one width that corresponds with a width of an interior space of the guide rod, the corresponding widths configured to prevent the pushrod from deflecting away from the hook assembly when the pushrod moves upward.

35. The umbrella of claim **29**, wherein the locking device comprises a spring configured to urge the hook assembly to engage the guide rod.

36. The umbrella of claim **29**, further comprising a counterweight system disposed within an interior space of the pole.

37. The umbrella of claim **36**, wherein the counterweight system comprises a ballast and a cord having a first end coupled with the ballast and a second end coupled with a lower hub.

38. The umbrella of claim **37**, wherein the pushrod is actuated by movement of the ballast.

39. An umbrella comprising: a pole; a shade assembly coupled with an upper portion of the pole, the shade assembly comprising: an upper hub coupled with a guide rod, and a lower hub moveable along the pole; and a locking device configured to engage the guide rod, the locking device comprising a pushrod positioned at least partially below the lower hub, wherein when closing the umbrella, the pushrod moves upward disengaging the locking device from the guide rod, allowing the lower hub to be moved downward.

40. The umbrella of claim **39**, wherein the locking device comprises a moveable locking surface configured to engage the guide rod.

41. The umbrella of claim **40**, wherein the locking device comprises a spring configured to move the moveable locking surface to engage the guide rod.

42. The umbrella of claim **40**, wherein the moveable locking surface comprises at least one prong configured to engage a corresponding recess of the guide rod when the umbrella is in an open state.

43. The umbrella of claim **39**, wherein the pushrod comprises a fixture configured to disengage the locking device when the pushrod moves upward.

44. The umbrella of claim **43**, wherein the fixture comprises an angled surface, the angled surface configured to disengage the locking device when the pushrod moves upward.

45. The umbrella of claim **39**, wherein the pushrod comprises at least one width that corresponds

with a width of an interior space of the guide rod, the corresponding widths configured to prevent the pushrod from deflecting away from the locking device when the pushrod moves upward.

46. The umbrella of claim 39, further comprising a counterweight system disposed within an interior space of the pole.

47. The umbrella of claim 46, wherein the counterweight system comprises a ballast and a cord having a first end coupled with the ballast and a second end coupled with the lower hub.

48. The umbrella of claim 47, wherein the pushrod is actuated by movement of the ballast.
